7 Work out

$$\sqrt{\frac{2.5 \times \sin 43°}{8.2^2 - 50.5}}$$

Give your answer correct to 3 significant figures.

$8.2^2 - 50.5 = 16.74$

$2.5 \sin 43° = 1.70499...$

$\sqrt{(1.70499... / 16.74)} = 0.319...$

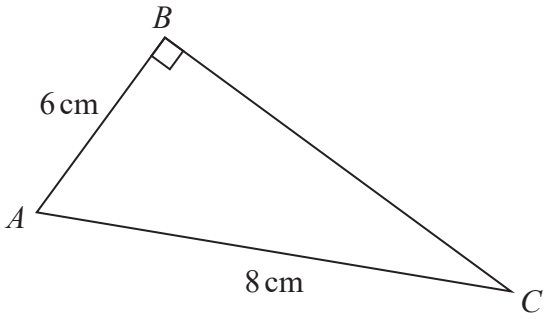
1 mark
Correct substitution and order.

1 mark
0.319 to 3 s.f.

0.319

(Total for Question 7 is 2 marks)

8 ABC is a right-angled triangle.



Here is Sarah's method to find the length of BC.

$$\begin{aligned} BC^2 &= AB^2 + AC^2 \\ &= 6^2 + 8^2 \\ &= 100 \\ BC &= 10 \end{aligned}$$

(a) What mistake has Sarah made in her method?

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(1)

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